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Economics and Conservation Tradeoffs

Week 9 Wednesday — Incentives, value, and who pays

NRES 315: Human Dimensions of Fish and Wildlife Management

Last updated: 2026-10-21

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One lesson for a shortened week

Monday was fall break. Today we connect Chapter 10's economic tools:

- What changes a person's incentive to act?
- Which benefits have no market price?
- Who gains, who pays, and what evidence would change our choice?

Source: Chapter 10, §§10.2–10.5; course calendar

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Economics asks about limited resources

Economics: how individuals and societies allocate limited resources to meet current and future needs.

A conservation choice uses land, time, labor, and funding. Its consequences can extend beyond the person making it.

Source: Chapter 10, §§10.2–10.3

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What you should be able to do today

- Diagnose an incentive or externality problem.
- Distinguish market price from economic value.
- Identify beneficiaries and cost bearers.
- Compare alternatives provisionally and propose deciding evidence.

Source: Chapter 10, §§10.3–10.5

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Property rights are a bundle

Rights can include **access, use, management, exclusion, and transfer**.

Who may enter, take a resource, change its management, exclude others, or transfer a right? These are different questions.

Source: Chapter 10, §10.3.1 and Table 10.1

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Classify the good before choosing the rule

Ask two questions: can people be **excluded**, and does one person's use **reduce what remains** for others?

- Private: excludable and rivalrous.
- Public: non-excludable and non-rivalrous.
- Common-pool: difficult to exclude, rivalrous.
- Club: excludable, largely non-rivalrous until congestion.

Source: Chapter 10, §10.3.2 and Table 10.2

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When price misses a consequence

An **externality** is a cost or benefit affecting others beyond a transaction.

Private costs and benefits can differ from social costs and benefits.

A price that omits those consequences can reward harmful activity or fail to reward beneficial activity.

Source: Chapter 10, §10.3.3

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Box 10.4: why offer cost-share?

A landowner pays to change a practice. Benefits can extend beyond the property.

Cost-share lowers part of the financial barrier to adoption. **Payments for ecosystem services** compensate for defined services.

Neither an offer nor a payment guarantees the intended outcome.

Source: Chapter 10, §10.3.3 and Box 10.4

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Pair check: trace the incentive

Using Box 10.4, name:

1. a cost borne by the landowner;
2. a benefit that extends beyond the property; and
3. how cost-share could change the choice.

Source: Chapter 10, Box 10.4

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Price does not capture every value

Market value concerns goods and services exchanged in markets.

Nonmarket value includes benefits and costs without a direct market price.

No price tag does not mean no value. A program payment does not measure its full public benefit.

Source: Chapter 10, §§10.4–10.4.1

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Five ways a resource can matter

- **Direct use:** fishing, viewing wildlife, hiking.
- **Indirect use:** ecosystem functions supporting other activities.
- **Option:** keeping the opportunity for future use.
- **Bequest:** conserving for future generations.
- **Existence:** knowing a species continues to exist.

Source: Chapter 10, §10.4.1

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Who receives the benefit and who pays?

An aggregate benefit can conceal uneven burdens.

Identify beneficiaries, cost bearers, and people missing from the accounting.

Compare installation, maintenance, foregone income, program costs, and benefits that extend beyond the parcel.

Source: Chapter 10, §10.5.4 and Box 10.4

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Match a valuation method to the question

Revealed preference: observe actual behavior, such as visits or property choices.

Stated preference: ask about choices or willingness to pay for specified outcomes; include values beyond current use.

Benefit transfer: apply an existing study to a comparable setting.

Source: Chapter 10, §§10.5.1–10.5.3

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Benefit-cost analysis needs a comparison

Compare changes against a **baseline** over a defined period.

Include market and nonmarket effects, identify who gains and pays, and state what cannot be measured.

Without estimates, we can identify needed evidence. We cannot calculate a net benefit or declare an economic winner.

Source: Chapter 10, §10.5.4

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Friday's decision: two restoration alternatives

Read the **Week 09 Case Brief: Comparing a Riparian Restoration Alternative**.

A: landowners choose to enroll; the partnership shares installation costs.

B: every eligible landowner would restore a buffer; authority and implementation capacity must first be confirmed.

Source: Friday Case Brief 09, §§2–5; concepts: Chapter 10, Box 10.4 and §§10.3.1–10.3.3

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Rehearse the comparison

With a partner, use Exhibits A and B to identify:

1. one incentive that differs between A and B;
2. one public benefit and one cost bearer; and
3. one information gap that could change your choice.

Source: Friday Case Brief 09, Exhibits A–B and §5; Chapter 10, Box 10.4 and §§10.4–10.5

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An evidence plan has a consequence

Name the information, how to obtain it, and whose information is needed.

Then state: **If we found ___, we would revise ___ because ___.**

A cost, enrollment, valuation, or authority assessment can be appropriate. Choose the one that tests the deciding uncertainty.

Source: Chapter 10, §§10.5.1–10.5.4; Friday Case Brief 09, §§5–7

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Before Friday and before submitting

Read the brief and find your **preassigned group and role**.

Submit the role record and **four numbered sections** by **10:50 a.m. Friday**.

Every present member completes **Friday Group Check 09**, including the submitter. The shared response is **80 points**; the private check is **20 points**.

Source: Friday Application 09 and Group Check 09

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A defensible choice names what could change it

Use economic concepts, case evidence, and the limits of both.

Recommend an alternative, explain its uneven consequences, and identify the finding that would make you reconsider.

Source: Chapter 10, §§10.5.4–10.6

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